ERRORS AND IMPLICIT HYPOTHESES IN CAMERON, GOETHALS, SEIDEL AND SHULT, *LINE GRAPHS, ROOT SYSTEMS, AND ELLIPTIC GEOMETRY*

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Scope

This document lists the errors in the printed paper [CGSS] (J. Algebra **43** (1976), 305–327) and the hypotheses that its statements leave implicit, as found while formalizing the paper. We checked every item against the published paper, and we quote the paper as printed; page numbers are those of the journal. The permanent record is `cgss/errata.md`. It also records 19 gaps in the arguments, which are neither errors nor missing hypotheses; we do not repeat them here, and we will cite them by their numbers there, as "Gap 14". Numbers in brackets, such as [15], refer to the references of [CGSS].

This version follows `cgss/errata.md`: 17 errors and 8 implicit hypotheses. A kernel-checked certificate in Lean proves the cospectral pair behind Errors 14, 16 and 17.

Each item ends with a *Status* line, which says where the correction now lives: its label in the blueprint (the structured proofs in the style of Lamport, in `blueprint/`) and its Lean declaration (in `LineGraphs/`), each with its file. Every numbered result of [CGSS] is in Lean, except Remark 2.11 (Shult's characterization, which is outside the scope); `dependency-diagram.md`, at the top of the repository, gives the declarations and their files. The Lean statements state the implicit hypotheses explicitly. The statement audit (`LineGraphs/AUDIT-statements.md`), which compared the final Lean with the paper itself, found no Lean statement that says something other than the paper without a recorded reason.

We keep the three graph conventions of the paper apart in our files: the *orthogonality graph* of Section 2 (adjacent at 90°), the *represented graph* of Section 4 (inner product 1), and the *obtuse graph* before Theorem 3.8 (inner product -1). The quotations keep the paper's G for a graph. In our own text graphs are X, Y, Z; we will write X_1 and X_2 for the paper's G_1 and G_2 in Section 5, and Y for the paper's graph H with $L(H) = G$ in Theorem 4.13. Strongly regular parameters are (n, k, a, c). The other symbols are the paper's, such as the parameters v, k, b, r, λ of a design.

Summary

In the column Status, a label is that of the blueprint, and "Lean" says that the corrected statement is in Lean; each item names the declarations.

Entry	Item of [CGSS]	Problem	Status
Error 1	Theorem 3.7	E_8: $(C - 18I)$ should be $(C - 28I)$	fixed: `thm:3.7`; Lean
Error 2	Example 4.6	coefficient $\frac{1}{3}$ should be $\frac{1}{2}$	fixed: `ex:4.6`; Lean
Error 3	Lemma 2.10	entry $(3, 1)$ of the block-sum matrix is 0, not 2	fixed: `lem:2.10`; Lean
Error 4	Theorem 5.8	$s \leq 2t - 1$ should be $s \geq 2t - 1$	fixed: `thm:5.8`; Lean
Error 5	Theorem 3.6	coclique of $T(6)$ should be clique	fixed: before `thm:3.6`; Lean (C5) avoids it

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Entry	Item of [CGSS]	Problem	Status
Error 6	Defs. 3.2, Thm. 2.13, proofs of 4.3, 4.4, 5.8	labels; Δ counted by lines	fixed: `def:3.2`, `thm:2.13`; Lean
Error 7	Theorem 4.13	the count $v-2$, $v-1$ is reversed	fixed: `lem:4-LG1`; Lean
Error 8	Theorem 4.4	"28 equiangular lines lie in $\mathbb{R}^7$" is false as stated	fixed: `thm:4.4`, `lem:4-EQ1`, `lem:4-EQ2`; Lean
Error 9	Section 6	$xAJ = -nJ$ and $AJ = -dJ$ fail in general; $x \mid n$ needs $n \geq 3$	fixed: `lem:6-fourth`; Lean
Error 10	Definition 3.9 (b)	S should be Σ	fixed: `def:3.9`; Lean
Error 11	Remark 3.11	four statements need qualification	fixed: `rem:3.11`, `thm:C`; Lean
Error 12	Lemma 2.12	"the unique strongly regular graphs on 9, 15, 27 vertices"	fixed: `lem:2.10`, `lem:2.12`; Lean
Error 13	several	further slips	fixed in the blueprint
Error 14	Remark 5.7	the eigenvalues for $\alpha = vw$; the sum of two multiplicities	fixed: `rem:5.7`; Lean
Error 15	Theorem 4.4	"$I + \frac{1}{2}A$ has rank ≤ 7" fails for $(n, d) = (28, 12)$	fixed: `thm:4.4`; Lean
Error 16	Theorem 5.8	false without $b \geq v$	fixed: `thm:5.8`; Lean
Error 17	Corollary 5.9	the exception must be unordered	fixed: `cor:5.9`; Lean
Impl. hyp. 1	Theorem 4.10; Section 1	X connected	fixed: `thm:4.10`, `lem:4-rootsystem`; Lean
Impl. hyp. 2	Theorem 4.13	a note on how we read "least eigenvalue -2"; nothing is missing	none needed: `thm:4.13`; Lean as printed
Impl. hyp. 3	Lemmas 5.4, 5.5	no isolated vertices; an edge	fixed: `lem:5.4`, `lem:5.5`; Lean
Impl. hyp. 4	Hypothesis 2.4	at least two lines	fixed: `hyp:2.4`, `lem:2.12`, `thm:2.13`, `thm:3.5`; Lean
Impl. hyp. 5	Lemma 5.6	X_2 without isolated vertices	fixed: `lem:5.6`; Lean
Impl. hyp. 6	Section 5	meaning of "characterized by its spectrum"	fixed: `def:flag`; Lean
Impl. hyp. 7	Definition 2.2	maximal within the span of the lines	fixed: `def:2.2`; Lean
Impl. hyp. 8	Theorem 3.12	either part of the partition may be empty	fixed: `thm:3.12`; Lean

1. Errors in the printed paper

Error 1. Theorem 3.7, the constant for E_8. *Where.* Theorem 3.7; p. 312.

Printed. "E_8: $(C + 2I)(C - 18I) = 0$".

Correct. $(C + 2I)(C - 28I) = 0$. The bound 28 on the valency in Theorem 4.3 is this constant (Gap 14); with the printed 18, the same argument would give 18, which the example in Theorem 4.3 contradicts.

Evidence. Take one vector of squared norm 2 on each of the 120 lines of E_8. These vectors form a tight frame in $\mathbb{R}^8$, $\sum \mathbf{v}\mathbf{v}^T = 30I$, so the Gram matrix $2I + C$ satisfies $(2I + C)^2 = 30(2I + C)$, and

C has eigenvalues 28 and -2, with multiplicities 8 and 112. Equivalently, each line of E_8 is at $60°$ to $56 = 2 \cdot 28$ others, and the trace argument that the paper gives for D_n yields $\gamma_1 + \cdots + \gamma_8 = 224$, so each γ_i is 28. We checked the eigenvalues numerically for E_6, E_7 and E_8; the printed values 10 and 16 for E_6 and E_7 are correct.

Repair. We derive all five constants from the frame identity. Let N be the number of lines and ℓ the rank. Over one vector per line, $\sum \mathbf{v}\mathbf{v}^T = \frac{2N}{\ell} I$ on the span of the roots (computation C6); in coordinates, this is $\frac{2N}{\ell}$ times the orthogonal projection onto the span, since the models of A_n, E_7, E_6 and the first model of E_8 do not span their coordinate spaces. The second constant is then $\frac{2N}{\ell} - 2$, which is 28 for E_8, where $\frac{N}{\ell} = 15$. The diagonal of C^2 is LN6 of `cgss/prerequisites.md` (computation C20). The same derivation is in `cgss/proofs/remark311/remark311.tex`.

Status. Fixed. Blueprint: Theorem 3.7, `thm:3.7`, through the frame identity within the span, `lem:sec3-frame`, and C6, `comp:C6` (`blueprint/sec3-rootsystems.tex`). Lean: `LineSystem.signedAdj_quadratic_E`$_8$, and the same name ending in `_An`, `_Dn`, `_E`$_7$ and `_E`$_6$ for the other types, all from `LineSystem.frame_identity` (`LineGraphs/Computations/C6.lean`); C6 is `Computations.C6.coordinate_frame_matrices`.

Error 2. Example 4.6, the Schläfli graph. *Where.* Example 4.6; p. 319.

Printed. "$\{\frac{1}{3}\Sigma e_k - e_i - e_7;\ i \in T(6,1)\}$".

Correct. $\frac{1}{2}\sum_{k=1}^{8} \mathbf{e}_k - \mathbf{e}_i - \mathbf{e}_7$, as in the third family.

Evidence. With $\frac{1}{3}$ these vectors have squared norm $14/9$, so they do not lie in the model (ii) of E_8. With $\frac{1}{2}$ the 27 vectors lie in E_8 and represent a strongly regular graph with parameters $(27, 16, 10, 8)$ (computation C9).

Repair. We use $\frac{1}{2}$; computation C9 checks the vectors and the parameters.

Status. Fixed. Blueprint: Example 4.6, `ex:4.6`, with C9, `comp:C9` (`blueprint/sec4-leasteigenvalue.tex`). Lean: `SimpleGraph.schlafli_isRepresentedIn_E`$_8$ and `SimpleGraph.schlafli_isSRGWith` (`LineGraphs/Computations/C9Schlafli.lean`).

Error 3. Lemma 2.10, the matrix of block sums. *Where.* Proof of Lemma 2.10; p. 309.

Printed. The third row of the 5×5 matrix of block sums is "$2 \quad 0 \quad k^2 + k \quad k^2 - k \quad k^2 - k$".

Correct. Its first entry is 0. The block of the first matrix that it sums, the inner products of the vertices of K with u, is 0, since the vertices of K are adjacent to u; as printed, the second matrix is not even symmetric.

Evidence. Symmetry, and a direct computation of the block sums of the first matrix. The determinant that the proof uses is that of the matrix with its first row and column deleted, which does not contain the wrong entry; it is $2k^5(5 - k)$, as stated. Thus the argument is unaffected.

Repair. The blueprint recomputes the block sums, with 0 in position $(3, 1)$; computation C1 is the determinant.

Status. Fixed. Blueprint: Lemma 2.10, `lem:2.10`, step `210gram`, which computes the block sums in its substeps `210blocks` and `210sums` and corrects the entry in its closing step, and C1, step `210det` (`comp:C1`), in `blueprint/sec2-lines.tex`. Lean: `Computations.C1.blockSumMatrix`, with the entry 0, is the Gram matrix of the five block sums by `LineSystem.Hyp24.gram_blockSumVectors` (`LineGraphs/LineSystem/OrthGraphGram.lean`); C1 is `Computations.C1.det_blockSumMinor` (`LineGraphs/Computations/C1.lean`).

Error 4. Theorem 5.8, the direction of an inequality. *Where.* Proof of Theorem 5.8; p. 323.

Printed. "$k \leqslant v$ implies $s \leqslant 2t-1$".

Correct. $s \geq 2t-1$, as in the statement, case (iii).

Evidence. Here $k = st$ and $v = \frac{1}{2}s(s+1)$ with $s, t > 0$ (the remark after Theorem 5.8), so $k \leq v$ if and only if $t \leq \frac{1}{2}(s+1)$, that is, $s \geq 2t-1$.

Repair. The blueprint of Theorem 5.8 proves $s \geq 2t-1$ from $k \leq v$; computation C11.

Status. Fixed. Blueprint: Theorem 5.8, `thm:5.8`, step `s558nbineq` (`blueprint/sec5-spectral.tex`). Lean: `Computations.C11.two_mul_sub_one_le` (`LineGraphs/Computations/C11.lean`), used by Theorem 5.8, `Design.flagGraph_cospectral` (`LineGraphs/LineGraphSpectra/Theorem58.lean`).

Error 5. Theorem 3.6, clique and coclique. *Where.* Proof of Theorem 3.6; p. 312.

Printed. "$A_7 \subset E_7$ since a 5-coclique is imbeddable in the triangular graph on 6 symbols".

Correct. The argument mixes the two graphs on Γ_a. In the graph of Section 2, where adjacency is orthogonality, Γ_a of A_7 is a coclique on 5 vertices, and Γ_a of E_7 is the complement of $T(6)$, with parameters $(15, 6, 1, 3)$. The inclusion needs a 5-coclique of the complement of $T(6)$, that is, a 5-clique of $T(6)$: the five pairs through a fixed symbol. $T(6)$ itself has no coclique on more than 3 vertices. The conclusion stands.

Evidence. The definitions; computation C5.

Repair. Computation C5 proves $A_7 \subset E_7$ by an explicit inclusion of coordinates, and does not need the 5-clique.

Status. Fixed. Blueprint: a remark just before Theorem 3.6 (`thm:3.6`) records the correction, and C5, `comp:C5`, proves the inclusion (`blueprint/sec3-rootsystems.tex`). Lean: `Computations.C5.inclusions_and_obstructions` (`LineGraphs/Computations/C5.lean`), which Theorem 3.6, `LineSystem.classification_of_isMaximal`, uses.

Error 6. Minor inconsistencies of labels and counts. *Where.* Definition 3.2 (p. 311), Theorem 2.13 (p. 310), the proofs of Theorems 4.3 and 4.4 (pp. 317–318) and of Theorem 5.8 (p. 323).

Printed. Definition 3.2 labels its second model "(i)"; later references call it (ii): "We use Definition 3.2(ii) for E_8" (p. 318). Theorem 2.13 gives "$|\Delta| = 36$", "$|\Delta| = 15$", "$|\Delta| = 6$". The proofs of Theorems 4.3 and 4.4 cite "Lemma 3.7" for Theorem 3.7, and the proof of Theorem 5.8 cites "Theorem 5.6" for Lemma 5.6.

Correct. The second model is (ii). The numbers of Theorem 2.13 count lines: as defined in Hypothesis 2.4, Δ is a set of vectors closed under negation, with 72, 30 and 12 elements. The line counts are the ones for which $3 + 3|\Gamma_a| + |\Delta|$ is 120, 63 and 36. The citations are of Theorem 3.7 and Lemma 5.6.

Evidence. The printed text.

Repair. The blueprint uses the correct labels and names the three models of E_8 (i), (ii), (iii). In Theorem 2.13 it writes $|\Delta|/2$ for the number of lines; in Lean, Δ is a set of vectors.

Status. Fixed. Blueprint: Definition 3.2, `def:3.2` (`blueprint/sec3-rootsystems.tex`); Theorem 2.13, `thm:2.13`, step `213exceptional` (`blueprint/sec2-lines.tex`); the citation of Theorem 3.7 for the valencies, in Lemma `lem:4-valency` (`blueprint/sec4-leasteigenvalue.tex`); and that of Lemma 5.6 in Theorem 5.8, `thm:5.8` (`blueprint/sec5-spectral.tex`). Lean: the three models are `LineSystem.E8viaA8`, `LineSystem.E8viaD8` and `LineSystem.E8viaSteiner` (`LineGraphs/LineSystem/Models.lean`), and `LineSystem.five_possibilities'` gives $|\Delta|/2 = 36, 15, 6$ (`LineGraphs/LineSystem/ClassificationModels.lean`).

Error 7. Theorem 4.13, a reversed count. *Where.* Proof of Theorem 4.13; p. 320.

Printed. "If $\{i,j\}$ and $\{i,k\}$ are edges of H, then the number of vertices of $L(H)$ adjacent to both is $v-2$ or $v-1$ according as $\{j,k\}$ is an edge or not."

Correct. It is $v-1$ if $\{j,k\}$ is an edge and $v-2$ otherwise.

Evidence. Here Y, the paper's H, is regular of valency v. The common neighbours of $\{i,j\}$ and $\{i,k\}$ in $L(Y)$ are the $v-2$ other edges on i, together with $\{j,k\}$ when it is an edge. The argument uses only that the two values differ, so it is unaffected.

Repair. The line graph case LG1 of `cgss/prerequisites.md`, §7.2, which the blueprint proves for Theorem 4.13, uses the corrected count.

Status. Fixed. Blueprint: Lemma (the line graph case, LG1), `lem:4-LG1`, used by Theorem 4.13, `thm:4.13` (`blueprint/sec4-leasteigenvalue.tex`). Lean: `SimpleGraph.lineGraph_isSRGWith_cases` (`LineGraphs/Seidel/LineGraphCases.lean`), which the proof of `SimpleGraph.seidel` uses.

Error 8. Theorem 4.4, 28 equiangular lines. *Where.* Proof of Theorem 4.4; p. 318.

Printed. "But any set of 28 equiangular lines lies in $\mathbb{R}^7$, hence the multiplicity of 0 as an eigenvalue actually is $\geqslant n-7$. This implies that $I+\frac{1}{2}A$ has rank $\leqslant 7$, hence the set of vectors we started with is contained in E_7 or D_7."

Correct. The first sentence is false as stated: 28 lines at angle $\arccos\frac{1}{3}$ can span $\mathbb{R}^{15}$. It is true for lines in $\mathbb{R}^m$ with $m \leq 14$, in particular in $\mathbb{R}^8$, which is the case used, but [15] (Lemmens and Seidel, *Equiangular lines*) does not state it (Gap 7). The last clause should also allow A_7 and the smaller systems; since $A_7 \subset E_7$, every indecomposable star-closed set of rank at most 7 lies in E_7 or D_7, and nothing changes.

Evidence. A single full pillar in the sense of [15] gives 28 lines at $\arccos\frac{1}{3}$ spanning $\mathbb{R}^{15}$ (`cgss/prerequisites.md`, §7.1).

Repair. EQ1 to EQ3 of `cgss/prerequisites.md`: the relative bound of [15], and the cone construction through Theorem 3.5 (decision 1 of `cgss/prerequisites.md`, §13).

Status. Fixed. Blueprint: Theorem 4.4, `thm:4.4`, through the relative bound EQ1, `lem:4-EQ1`, and the lines spanning eight dimensions, EQ2, `lem:4-EQ2` (`blueprint/sec4-leasteigenvalue.tex`). Lean: `InnerProductSpace.card_le_relativeBound` (`LineGraphs/Representation/Theorem44.lean`) and `InnerProductSpace.card_le_fourteen_of_finrank_eq_eight` (`LineGraphs/Spectral/RelativeBoundFourteen.lean`); Theorem 4.4 is `SimpleGraph.IsRepresentedIn.card_le_28_of_regular` and `SimpleGraph.IsRepresentedIn.degree_le_16_of_regular` (`LineGraphs/Representation/Theorem44Unconditional.lean`).

Error 9. Section 6, the row sums of A. *Where.* Section 6, before Theorem 6.2; p. 326.

Printed. "$(xA+nI)(xA-\frac{1}{2}n(n-2)I)=0$, $xAJ=-nJ$, hence x divides n" and "$(A+dI)(A-\frac{1}{2}d(n-2)I)=0$, $AJ=-dJ$".

Correct. The quadratic equations hold; the equations $xAJ=-nJ$ and $AJ=-dJ$ do not hold in general. Since $PP^T=nI+xA$, the equation $xAJ=-nJ$ says that $PP^T\mathbf{j}=0$, that is, $P^T\mathbf{j}=0$. The entry of $P^T\mathbf{j}$ at a column c is $\frac{1}{2}(\sigma(c)^2-n)$, where $\sigma(c)$ is the sum of column c of H_n. Thus $xAJ=-nJ$ holds if and only if every column sum of H_n is $\pm\sqrt{n}$. This depends on the signs of the rows of H_n, and it fails for every choice of signs unless n is a square. The conclusion that x divides n is true for another reason: as the paper says next, $d^2=(n/x)^2$ is the number of fourth rows that agree with three given rows in $\frac{1}{8}(n\pm x)$ places, so n/x is a rational number whose square is an integer. (For a fixed row i, the $n-1$ vectors $H_i\circ H_j$ form an orthogonal basis of $\mathbf{j}^\perp$, and

expanding $H_k \circ H_l$ in this basis gives the count.) The claim needs $n \geq 3$. For $n = 1$ and $n = 2$ the matrix has fewer than four rows, so Hypothesis 6.1 holds for every x, and x need not divide n (take $n = 2$ and $x = 3$); there is no Hadamard matrix of order 3. The assumption $x > 0$ costs nothing: for $n \geq 4$ and $x = 0$ we would have $PP^T = nI$, of rank $\binom{n}{2} > n$, whereas P has only n columns.

Evidence. For the Paley matrix H_{12}, the row sums of xA are -60 and 36, and no choice of row signs makes every column sum $\pm\sqrt{12}$. For the Sylvester matrix of order 8, every row sum of xA is $24 = \frac{1}{2}n(n-2)$, so $\mathbf{j}$ is an eigenvector of PP^T for $\frac{1}{2}n^2$, not for 0 (NumPy, scratch). The same check confirms Hypothesis 6.1 for H_{12} with $x = 4$. Theorem 6.2 is unaffected, since its proof uses only the quadratic equation, the fact that the lines lie in $\mathbf{j}^\perp$, and the orthogonality of the lines through a fixed row.

Repair. The blueprint states the quadratic equation, and derives $x \mid n$ from the count of fourth rows; computation C16. A shorter route: for $n > 2$ the eigenvalue $-n/x$ of the integer matrix C_H (the paper's A) has multiplicity $\frac{1}{2}(n-1)(n-2) > 0$, so it is an integer by the rational root theorem (`cgss/proofs/spectraltools/spectraltools.tex`, Theorem 9.6, used in its Theorem 10.4).

Status. Fixed. Blueprint: the background of Section 6, the quadratic equation, `lem:6-quadratic`, and the fourth rows, `lem:6-fourth`, with the remark after it (`blueprint/sec6-hadamard.tex`). Lean: `Matrix.IsHadamard.hadamardSignedAdj_quadratic` (`LineGraphs/Hadamard/Agreement.lean`) and `Matrix.IsHadamard.FourRowAgreement.dvd_card` (`LineGraphs/Hadamard/FourthRows.lean`), which assumes $n \geq 3$ and $x > 0$. The two false identities have no Lean statement; the statement audit records them as rightly absent.

Error 10. Definition 3.9 (b), S for Σ. *Where.* Definition 3.9 (b); p. 314.

Printed. "$\forall_{r\in\Sigma}\forall_{\lambda\in\mathbb{R}}((\lambda r \in S) \Rightarrow (\lambda \in \{1,-1\}))$".

Correct. $\lambda r \in \Sigma$. The letter S names the set of lines of Section 2.

Evidence. The printed text, and the sense of the definition. Read with Σ, condition (b) also excludes 0 from Σ (if $0 \in \Sigma$, then $\lambda 0 \in \Sigma$ for $\lambda = 2$).

Repair. Section 2 of `cgss/proofs/remark311/remark311.tex`, and item 2 of its section "Interpretations and corrections of the sources".

Status. Fixed. Blueprint: Definition 3.9, `def:3.9` (`blueprint/sec3-remark311.tex`). Lean: `IsRootSet` (`LineGraphs/RootSystems/Defs.lean`), whose field `reduced` is condition (b), read with Σ.

Error 11. Remark 3.11, statements that need qualification. *Where.* Remark 3.11; p. 314.

Printed. "Let $r, s \in \Sigma$ be not perpendicular. Since $4(r,s)^2/(r,r)(s,s)$ is a positive integer < 4, ..."; "the subset of a Root System consisting of vectors of any fixed length is itself a Root System, necessarily one of those described by Theorem 3.10"; "B_n and C_n: a set of n^2 lines in $\mathbb{R}^n$ with angles 90°, 60°, 45°, and equation $(C + 2I)(C - 2(n-1)I) = 0$", and the similar lines for F_4 and G_2.

Correct. (1) The roots r and s must also be non-proportional; for $s = \pm r$ the quotient is 4. (2) The roots of one length form a Root System in their own span, which need not be the whole space and need not be indecomposable: the long roots of B_2 form $A_1 \oplus A_1$, and the roots of length 1 in $\{\pm\mathbf{u}_1, \pm 2\mathbf{u}_2\}$ span a line. (3) The matrix C is defined before Theorem 3.7 only for vectors of one length. For two root lengths we take unit vectors on the lines, as the columns of a matrix U, and put $C = 2U^TU - 2I$; then the three printed equations hold. (4) The listed angles are the angles allowed: for $B_2 = C_2$ the angle 60° does not occur.

Evidence. `cgss/proofs/remark311/remark311.tex`, § "Interpretations and corrections of the sources", items 3 to 6, and its Lemma "angles" and Theorem C, with their checks.

Repair. `cgss/proofs/remark311/remark311.tex` (decision 5 of `cgss/prerequisites.md`, §13); the blueprint of Remark 3.11 follows it.

Status. Fixed. Blueprint: Remark 3.11, `rem:3.11`, through Lemma (angles and fixed-length subsystems), `lem:angles`, and Theorem C, `thm:C` (`blueprint/sec3-remark311.tex`). Lean: `IsRootSet.remark_lines` (`LineGraphs/RootSetClassification/RemarkLines.lean`), with `RootSetModel.Bn_Cn_same_lines` and the equations `RootSetModel.signedAdj_quadratic_Bn`, `_Cn`, `_F`$_4$ and `_G`$_2$; the subsets of one length are `IsRootSet.fixedLength_isRootSet`.

Error 12. Lemma 2.12, "the unique strongly regular graphs on 9, 15, 27 vertices". *Where.* Lemma 2.12; p. 310.

Printed. "…the void graph on n vertices, the graph consisting of n triangles meeting at one vertex, the unique strongly regular graphs on 9, 15, 27 vertices."

Correct. There are other strongly regular graphs on these numbers of vertices: $T(6)$ on 15 and the Schläfli graph on 27 (the complements of the intended graphs), and the degenerate $3K_3$ and $K_{3,3,3}$ on 9. The intended graphs are those of Lemma 2.10, which are strongly regular with parameters $(9,4,1,2)$, $(15,6,1,3)$ and $(27,10,1,5)$. The paper proves only that they are regular. Strong regularity takes two further observations, in the notation of Lemma 2.10: adjacent vertices have exactly one common neighbour, the third vertex of their triangle; and nonadjacent vertices u, v have the k common neighbours in K.

Evidence. $T(6)$ has parameters $(15,8,4,4)$ and the Schläfli graph $(27,16,10,8)$.

Repair. The blueprint names the graphs by their parameters, or as the collinearity graphs of the generalized quadrangles of orders $(2,1)$, $(2,2)$, $(2,4)$ (GQ1 and GQ2 of `cgss/prerequisites.md`).

Status. Fixed. Blueprint: Lemma 2.10, `lem:2.10`, proves strong regularity with parameters $(6k-3,2k,1,k)$, and Lemma 2.12, `lem:2.12`, lists the graphs by their parameters (`blueprint/sec2-lines.tex`). Lean: `LineSystem.Hyp24.orthGraph_isSRGWith` (`LineGraphs/LineSystem/OrthGraphParameters.lean`) and `LineSystem.Hyp24.orthGraph_cases` (`LineGraphs/LineSystem/OrthGraphCases.lean`).

Error 13. Further slips. *Where.* Various pages.

Printed. (1) Lemma 3.4: "$|A| = \frac{1}{2}n(n+1)$" for $|A_n|$ (p. 311). (2) Definition 3.2 (ii): "$\epsilon_i = \pm 1$" appears twice in the display (p. 311). (3) Theorem 3.12: "two lines in different subsets are at angles 90° and 45°" for "90° or 45°" (p. 315). (4) Section 4: "is and orthonormal basis" (p. 315); "by delecting a one-factor" (p. 316); "whose lines from a subset of D_n" (p. 316); and Corollary 5.10: "The graphs ar switching equivalent" (p. 324). (5) The representation of the generalized line graphs, and the proof of Theorem 4.2: "$1 \leqslant \alpha \leqslant \alpha_i$" for $1 \leq \alpha \leq a_i$ (pp. 316 and 317). (6) Proof of Theorem 4.4: "The projections onto the lines spanned by these vectors are elements of $\mathbb{R}^{28}$ and have Gram matrix $(1-\frac{d+2}{2n})^2 I + \cdots$" (p. 317). (7) Proof of Theorem 4.4: "for each eigenvalue $\alpha \neq d$ of A" (p. 318). (8) Remark 5.7: "two eigenvalues $\delta := \gamma - \frac{1}{2}(v+w)$ and $\delta := \gamma' - \frac{1}{2}(v+w)$" (p. 322).

Correct. Items (1) to (5) are misprints with evident corrections. In (6), the vectors have squared norm $1-\frac{d+2}{2n}$, and the displayed matrix is the Gram matrix of the rank-one matrices $\mathbf{p}\mathbf{p}^T$, not of the projections, whose Gram matrix is the displayed one divided by $(1-\frac{d+2}{2n})^2$; the conclusion is the same. In (7), read "for each eigenvalue α of A with an eigenvector orthogonal to $\mathbf{j}$". The two readings differ only when the graph is disconnected; then d is also an eigenvalue on $\mathbf{j}^\perp$, the inequality holds for it, and the sum over $\alpha_1,\ldots,\alpha_7$ needs it. In (8), the second is $\delta' := \gamma' - \frac{1}{2}(v+w)$.

Evidence. The printed text. None of these affects an argument.

Repair. The blueprint uses the corrected forms.

Status. Fixed in the blueprint: (1) Lemma 3.4, `lem:3.4`; (2) Definition 3.2, `def:3.2`; (3) Theorem 3.12, `thm:3.12`, whose Lean statement, `LineSystem.subset_rootSet_iff_bipartition`, also reads "or"; (4) Theorem 4.2, `thm:4.2`, for "form"; (5) Lemma (the representation of a generalized line graph), `lem:4-glgrep`; (6) and (7) Theorem 4.4, `thm:4.4`; (8) Remark 5.7, `rem:5.7`. The Lean statements are those of the corrected forms; for instance, `LineSystem.numLines_An` counts the lines of A_n.

Error 14. Remark 5.7, the eigenvalues of $A(X_2)$. *Where.* Remark 5.7; pp. 322–323.

Printed. "To each eigenvalue $\alpha \neq 0$ of $B_1 B_1^T$ there correspond ... two eigenvalues $\delta := \gamma - \frac{1}{2}(v+w)$ and $\delta := \gamma' - \frac{1}{2}(v+w)$ of $A(G_2)$" (the second δ is the misprint of Error 13, item 8) and "For an eigenvalue 0 of $A(G_1)$ the corresponding eigenvalues of $A(G_2)$ are $\pm\frac{1}{2}(v-w)$. Their multiplicities need not be equal, add up to $q-p$".

Correct. In the paper's notation, the parts of X_1 have sizes $p \leq q$ and valencies v and w, and $\rho = \operatorname{rk} B_1$. (a) For $\alpha = vw$ the roots of $(\gamma - v)(\gamma - w) = \alpha$ are $v + w$ and 0, and 0 is not an eigenvalue of $M_2 M_2^T$, which is nonsingular; thus only $\frac{1}{2}(v+w)$ occurs, not $-\frac{1}{2}(v+w)$. (b) The multiplicities of $\pm\frac{1}{2}(v-w)$ are $p-\rho$ and $q-\rho$; their *difference* is $q-p$, and their sum is $p+q-2\rho$.

Evidence. Take $X_1 = K_{6,10}$ and $X_2 = T(6)$. Then $L(K_{6,10})$ and $L(T(6))$ both have spectrum $14^1\, 8^5\, 4^9\, (-2)^{45}$, X_2 is 8-regular and not bipartite, and $|X_1| = |X_2| + 1$; the eigenvalues ± 2 of $T(6)$ have multiplicities 5 and 9, which add up to $14 \neq q - p = 4$. This is the situation of Corollary 5.9 and Remark 5.12, where $B_1 = J$ has rank 1. We checked it by hand, and a kernel-checked certificate also proves it (see the Status). Theorem 5.8 does not use these statements.

Repair. The spectral toolkit, Corollary 6.8 of `cgss/proofs/spectraltools/spectraltools.tex`, states the correct spectrum, and the blueprint of Remark 5.7 uses it.

Status. Fixed. Blueprint: Remark 5.7, `rem:5.7` (`blueprint/sec5-spectral.tex`), which gives the spectrum of $A(X_2)$, with the multiplicities $p-\rho$ and $q-\rho$. Lean: `SimpleGraph.lineGraph_cospectral_bipartite_spectrum_of_not_isBipartite` (`LineGraphs/LineGraphSpectra/Nonbipartite.lean`). The evidence is kernel-checked: the spectra of $L(K_{10,6})$, the flag graph of the trivial design $(6, 6, 10, 10, 10)$, of $L(L(K_6)) = L(T(6))$ and of $T(6)$ are `Computations.TrivialDesignException.spectrum_flagGraph`, `.spectrum_double_lineGraph` and `.spectrum_triangular`, and the cospectrality is `.flagGraph_cospectral_double_lineGraph` (`LineGraphs/Computations/TrivialDesignException.lean`).

Error 15. Theorem 4.4, the rank of $I + \frac{1}{2}A$ when $d = 12$. *Where.* Proof of Theorem 4.4; p. 318.

Printed. The matrix $I + \frac{2}{3}A - \frac{1}{3}(J - I)$ "has the simple eigenvalue $1 + \frac{2}{3}d - \frac{1}{3}(n-1) > 0$", and later "This implies that $I + \frac{1}{2}A$ has rank $\leqslant 7$, hence the set of vectors we started with is contained in E_7 or D_7."

Correct. For $n = 28$ the eigenvalue on $\mathbf{j}$ is $\frac{1}{3}(2d - 24)$, which is 0 when $d = 12$, and $(n, d) = (28, 12)$ is the case of $T(8)$ and the Chang graphs, which lie in E_8. For $T(8)$, with spectrum $12^1\, 4^7\, (-2)^{20}$, the matrix $I + \frac{2}{3}A - \frac{1}{3}(J - I)$ has eigenvalues 0, 4^7, 0^{20}, and so rank 7, while $I + \frac{1}{2}A$ has eigenvalues 7, 3^7, 0^{20}, and so rank 8. Thus the sentence is false for $d = 12$. The conclusion $d \leq 16$ stands, since $12 \leq 16$.

Evidence. We checked it by hand from the spectrum of $T(8)$.

Repair. The blueprint of Theorem 4.4 splits into $d \leq 12$, where there is nothing to prove for the bound, and $d > 12$, where the eigenvalue on $\mathbf{j}$ is positive and the rank argument goes through.

Status. Fixed. Blueprint: Theorem 4.4, `thm:4.4`, step `s44rankE`, with the case $d \leq 12$ in step `s44big` (`blueprint/sec4-leasteigenvalue.tex`). Lean:

`SimpleGraph.IsRepresentedIn.degree_le_16_of_regular` (`LineGraphs/Representation/Theorem44Unconditional.lean`); that the case $(28, 12)$ occurs is part of `SimpleGraph.regular_28_12_and_27_16_isRepresentedIn_E`$_8$ (`LineGraphs/Representation/Theorem44.lean`).

Error 16. Theorem 5.8 needs $b \geq v$. *Where.* Theorem 5.8 and the remark after it; pp. 323–324.

Printed. "Let D_1 be a 2-design with parameters $v, k, b, r, \lambda \ldots$ Then one of the following holds", with the cases (i) "$v + b \leqslant 9, \ldots$", (ii) "$G_2 = F(D_2), \ldots$" and (iii) "$v = \frac{1}{2}s(s+1)$, $k = st$, $b = \frac{1}{2}(s+1)(s+2), \ldots$".

Correct. The statement fails without $b \geq v$. Take the trivial design $v = k = 10$, $b = r = \lambda = 6$: its flag graph $L(K_{10,6})$ is cospectral with $L(T(6))$ (Error 14). Case (i) fails, since $v + b = 16$; (ii) fails, since the only design with these parameters has flag graph $L(K_{10,6})$, and $T(6)$ is not bipartite; and (iii) fails, since $v = 10$ forces $s = 4$, and then $b = 15$. The dual design ($v = 6$, $b = 10$) satisfies (iii) with $s = 3$, $t = 2$. The paper's remark after the theorem ($s, t > 0$ when $\lambda < r$; otherwise pass to the dual) is the intended hypothesis, and every corollary meets it.

Evidence. We checked it by hand from the spectra in Error 14, and a kernel-checked certificate proves that $L(K_{10,6})$ and $L(T(6)) = L(L(K_6))$ are cospectral and not isomorphic.

Repair. The blueprint of Theorem 5.8 assumes $b \geq v$.

Status. Fixed. Blueprint: Theorem 5.8, `thm:5.8` (`blueprint/sec5-spectral.tex`). Lean: `Design.flagGraph_cospectral`, with the hypothesis `hvb` that $v \leq b$ (`LineGraphs/LineGraphSpectra/Theorem58.lean`). The evidence: `Computations.TrivialDesignException.lineGraph_completeBipartite_cospectral_double_lineGraph` and `.flagGraph_not_nonempty_iso_double_lineGraph`, the latter for the flag graph of the dual design $(6, 6, 10, 10, 10)$, which is $L(K_{10,6})$ again (`LineGraphs/Computations/TrivialDesignException.lean`).

Error 17. Corollary 5.9, the exception is unordered. *Where.* Corollary 5.9; p. 324.

Printed. "the graph $L(K_{m,n})$ is characterized by its spectrum, unless $m = 2t^2 + t$, $n = 2t^2 - t$, and there exists a symmetric Hadamard matrix with constant diagonal of order $4t^2$."

Correct. The exception must read $\{m, n\} = \{2t^2 + t, 2t^2 - t\}$: for $(m, n) = (6, 10)$ the printed condition does not apply, but $L(K_{6,10}) = L(K_{10,6})$ is not characterized (it is cospectral with $L(T(6))$; $t = 2$).

Evidence. The pair of Error 14. A kernel-checked certificate proves that the flag graph $L(K_{10,6})$ of the trivial design $(6, 6, 10, 10, 10)$ is cospectral with $L(L(K_6)) = L(T(6))$, is not isomorphic to it, and is not characterized by its spectrum in the sense of Implicit hypothesis 6.

Repair. The blueprint states Corollary 5.9 for the unordered pair $\{m, n\}$, and reduces it to $m \geq n$ by symmetry.

Status. Fixed. Blueprint: Corollary 5.9, `cor:5.9`, step `s559ordered` (`blueprint/sec5-spectral.tex`). Lean: `SimpleGraph.lineGraph_completeBipartite_characterized`, whose exception `SimpleGraph.HasCompleteBipartiteHadamardException` is symmetric in m and n, through `SimpleGraph.lineGraph_completeBipartite_characterized_of_le` (`LineGraphs/LineGraphSpectra/Corollaries.lean`). The evidence: `Computations.TrivialDesignException.flagGraph_cospectral_double_lineGraph`, `.flagGraph_not_nonempty_iso_double_lineGraph` and `.not_flagGraphCharacterized` (`LineGraphs/Computations/TrivialDesignException.lean`).

2. Implicit hypotheses

The Lean statements state each of these explicitly. Implicit hypothesis 2 is a note on how we read a phrase of the paper, not a hypothesis that the paper leaves out; it keeps its number, which other files cite.

Implicit hypothesis 1. Theorem 4.10 needs X to be connected. *Where.* Theorem 4.10; p. 319. The claim of the introduction that the lines of such a graph lie in one of the root systems needs the condition too; p. 306.

Printed. "If G is a graph with least eigenvalue -2, then either (i) G is a generalized line graph; or (ii) G is represented by a subset of E_8." In the introduction: "Any graph with least eigenvalue -2 may be viewed as a set of lines at 60° and 90°. As a consequence of the main theorem, this set is contained in one of the root systems."

Missing. X is connected.

Evidence. Two disjoint copies of the Petersen graph have least eigenvalue -2 and do not form a generalized line graph. Their matrix $2I + A$ has rank $12 > 8$, so they are not represented in E_8 either. Thus their lines lie in none of the root systems A_n, D_n, E_6, E_7, E_8: subsets of A_n and D_n represent generalized line graphs (Theorems 4.1 and 4.2), and E_6, E_7 and E_8 span at most 8 dimensions. Section 4 states the condition (p. 315).

Repair. The Lean statement of Theorem 4.10 assumes that X is connected, and so does the Lean form of the claim of the introduction.

Status. Fixed. Blueprint: Theorem 4.10, `thm:4.10`, and Lemma (a root system containing the lines), `lem:4-rootsystem` (`blueprint/sec4-leasteigenvalue.tex`). Lean: `SimpleGraph.hoffman` (`LineGraphs/Representation/Hoffman.lean`) and `SimpleGraph.IsRepresentation.exists_isRepresentedIn_rootType` (`LineGraphs/Representation/RootTypeClassified.lean`), both with the hypothesis `X.Connected`.

Implicit hypothesis 2. Theorem 4.13 and "least eigenvalue -2". *Where.* Theorem 4.13; p. 320. The phrase also occurs in Theorems 4.10 and 4.11 (pp. 319 and 320).

Printed. "If G is a connected strongly regular graph with least eigenvalue -2, then either ..."

Note. Nothing is missing from the printed theorem. Read literally, "least eigenvalue -2" says that the least eigenvalue is -2, so that -2 is an eigenvalue, and the paper uses the phrase in this sense. Thus Theorem 4.13 is correct as printed. In Theorems 4.10 and 4.11 we read the phrase as the lower bound $\lambda_{\min} \geq -2$, that is, $2I + A \succeq 0$, which makes these two theorems stronger. Theorem 4.13 is false under that reading.

Evidence. The paper writes "the least eigenvalue of $L(G; a_1, \ldots, a_m)$ is -2 if the number of vertices exceeds n" (p. 316), and the proof of Theorem 4.13 ends with "$K_n = L(K_{1,n})$ and $C_5 = L(C_5)$ do not have -2 as an eigenvalue". Under the lower bound reading, K_n with $n > 36$ is a counterexample: it is strongly regular in Mathlib's sense (`IsSRGWith`) and has least eigenvalue -1, but it is none of the graphs in (i) or (ii).

Repair. None is needed. The Lean statement assumes the lower bound and, as a second hypothesis, that -2 is an eigenvalue; together these say that the least eigenvalue is -2, by the bridge lemma `SimpleGraph.leastEigenvalue_eq_neg_two_iff`, so the Lean states the printed theorem.

Status. Nothing to fix. Blueprint: Theorem 4.13, `thm:4.13`, with its two hypotheses (`blueprint/sec4-leasteigenvalue.tex`). Lean: `SimpleGraph.seidel` (`LineGraphs/Representation/Seidel413.lean`), whose hypotheses `hp` and `hneg` say that $2I + A \succeq 0$ and that -2 is an eigenvalue; the bridge is `SimpleGraph.leastEigenvalue_eq_neg_two_iff` (`LineGraphs/Spectral/Positivity.lean`).

Theorems 4.10 and 4.11, `SimpleGraph.hoffman` and `SimpleGraph.hoffman_rayChaudhuri`, assume only the lower bound.

Implicit hypothesis 3. Lemmas 5.4 and 5.5. *Where.* Lemmas 5.4 and 5.5; p. 321.

Printed. Lemma 5.4: "If $L(G)$ is regular and connected, then either G is regular, or G is bipartite and the vertices in the same bipartite block have the same valency." Lemma 5.5: "The spectrum of a regular graph is determined by the spectrum of its line graph."

Missing. Lemma 5.4 needs X to have no isolated vertices, and Lemma 5.5 needs at least one edge.

Evidence. K_2 together with an isolated vertex is a counterexample to Lemma 5.4. All edgeless graphs have the same line graph, so Lemma 5.5 fails without an edge.

Repair. SP4 of `cgss/prerequisites.md`, and the blueprint of Lemmas 5.4 and 5.5, assume these.

Status. Fixed. Blueprint: Lemma 5.4, `lem:5.4`, and Lemma 5.5, `lem:5.5` (`blueprint/sec5-spectral.tex`). Lean: `SimpleGraph.regular_or_isBiregularWith_of_lineGraph_regular` (`LineGraphs/Spectral/LineGraphRegular.lean`) assumes that every vertex has positive degree, and `SimpleGraph.cospectral_of_lineGraph_cospectral` (`LineGraphs/Spectral/LineGraphRecovery.lean`) assumes that both graphs have an edge.

Implicit hypothesis 4. Hypothesis 2.4 needs two lines. *Where.* After Hypothesis 2.4; p. 307.

Printed. "Without loss of generality we assume that S contains lines $\langle a\rangle$ and $\langle b\rangle$ whose spanning vectors a and b have the inner product $(a, b) = -1$."

Missing. S has at least two lines; in particular, S is not empty.

Evidence. The assumption is possible exactly when S has at least two lines: an indecomposable set with two or more lines contains two lines at $60°$, since otherwise all its lines would be mutually orthogonal, and a choice of signs gives $(\mathbf{a}, \mathbf{b}) = -1$. A single line is an indecomposable star-closed set, namely A_1, and for it Γ_a, Δ and Σ are not defined. Thus Lemma 2.12, Theorem 2.13 and Theorem 3.5 need the case of one line separately; the first line of Theorem 2.13 with $n = 1$ is this case. The empty set must be excluded too.

Repair. The blueprint of Theorems 2.13 and 3.5 splits on the number of lines. In Lean the structure `LineSystem.Hyp24` carries **a** and **b**, and the classification, `LineSystem.classification`, assumes a nonempty set (`hne`).

Status. Fixed. Blueprint: Hypothesis 2.4, `hyp:2.4`, Lemma 2.12, `lem:2.12`, step `212star`, and Theorem 2.13, `thm:2.13` (`blueprint/sec2-lines.tex`); Theorem 3.5, `thm:3.5` (`blueprint/sec3-rootsystems.tex`). Lean: `LineSystem.Hyp24` (`LineGraphs/LineSystem/Hyp24.lean`); the split `LineSystem.IsIndecomposable.eq_pair_or_exists_hyp24` (`LineGraphs/LineSystem/OrthGraphCases.lean`); and `LineSystem.five_possibilities'` and `LineSystem.classification`, which assume that the set of roots is nonempty.

Implicit hypothesis 5. Lemma 5.6 needs X_2 without isolated vertices. *Where.* Lemma 5.6; p. 321.

Printed. Lemma 5.6: "Let G_1 be connected bipartite such that $L(G_1)$ is regular. Let G_2 be such that $\operatorname{spec}(L(G_1)) = \operatorname{spec}(L(G_2))$. Then either (i) ... or (ii) G_1 is not regular; G_2 is regular, not bipartite; $|G_1| = |G_2| + 1$."

Missing. X_2, the paper's G_2, has no isolated vertices.

Evidence. Take $X_1 = K_{1,3}$, and let X_2 be a triangle together with an isolated vertex. Then $L(X_1) = L(X_2) = K_3$, but X_2 is neither bipartite nor regular. The proof applies Lemma 5.4 to X_2, which needs no isolated vertices (Implicit hypothesis 3); it also uses that $L(X_2)$ is regular and

connected, which follows from Lemmas 5.1 and 5.2. Theorem 5.8 is unaffected, since there the graph Y with $X_2 = L(Y)$ can be chosen without isolated vertices.

Repair. The blueprint and the Lean statement of Lemma 5.6 assume that X_2 has no isolated vertices.

Status. Fixed. Blueprint: Lemma 5.6, `lem:5.6` (`blueprint/sec5-spectral.tex`). Lean: `SimpleGraph.lineGraph_cospectral_bipartite_forall_isBiregularWith` (`LineGraphs/LineGraphSpectra/Bipartite.lean`), whose hypothesis `hnoY` says that every vertex of the second graph has positive degree.

Implicit hypothesis 6. The meaning of "characterized by its spectrum". *Where.* Corollaries 5.9–5.11, Remark 5.12 and Example 5.13; pp. 324–325.

Printed. For instance, "The flag graph of a 2-design with $\lambda = 1$ is characterized by its spectrum."

Missing. The paper does not define the phrase. Its proofs show, through Theorem 5.8 (ii), that every graph with the spectrum of $F(D_1)$ is $F(D_2)$ for some 2-design D_2 with the same parameters, and this is the sense of Hoffman and Ray-Chaudhuri [14] and of Doob [6, 7].

Evidence. Read as "determined up to isomorphism by its spectrum", Corollaries 5.10 and 5.11 and Remark 5.12 are false. The spectrum of $F(D)$ depends only on the parameters, since $NN^T = (r - \lambda)I + \lambda J$ (proof of Theorem 5.8), whereas $F(D) = L(I(D))$ determines $I(D)$ by Whitney's theorem, and so determines D up to duality. Thus the two Steiner triple systems on 13 points have cospectral, non-isomorphic flag graphs (against Corollary 5.11), and the five symmetric designs with parameters $(15, 7, 3)$ give at least three (against Corollary 5.10 and Remark 5.12, with $\lambda = 3$). For Corollary 5.9 the two readings agree, since the trivial design with given parameters is unique.

Repair. The blueprint and the Lean statements define the phrase in this sense: every graph with the spectrum of $F(D_1)$ is isomorphic to $F(D_2)$ for some 2-design D_2 with the parameters of D_1. On the paper's route, the "not characterized" of Example 5.13 is then EX4 of `cgss/prerequisites.md`, a case of Whitney's theorem; the Lean proof avoids EX4 by a clique bound (blueprint, Example 5.13).

Status. Fixed. Blueprint: Definition (incidence graphs, flag graphs, characterization), `def:flag`, used from Corollary 5.9, `cor:5.9`, to Example 5.13, `ex:5.13` (`blueprint/sec5-spectral.tex`). Lean: `Design.FlagGraphCharacterized` (`LineGraphs/LineGraphSpectra/DesignGraphs.lean`), used by `Design.IsSymmetric.flagGraphCharacterized`, `Design.flagGraphCharacterized_of_lam_eq_one`, `Design.flagGraphCharacterized_of_lam_lt_nineteen` and `Design.not_flagGraphCharacterized_of_resolvable`, the last for Example 5.13.

Implicit hypothesis 7. Definition 2.2 means maximal within the span. *Where.* Section 2, first paragraph, and Definition 2.2; p. 306.

Printed. "Without loss of generality the lines are taken to pass through the origin of $\mathbb{R}^n$ and to span $\mathbb{R}^n$." and "A set of lines at 60° and 90° in $\mathbb{R}^n$ is maximal if any line of $\mathbb{R}^n$ belongs to the set whenever it has angles 60° or 90° with all lines of the set."

Missing. Maximality is meant relative to the space that the lines span.

Evidence. Theorem 3.6 needs this reading: it calls A_7 and A_8 non-maximal because $A_7 \subset E_7$ and $A_8 \subset E_8$ in the same dimension, and A_n is defined in Definition 3.1 in $\mathbb{R}^{n+1}$, where $\mathbf{e}_1 + \mathbf{e}_2$ is at 60° or 90° to all its lines, so that no A_n would be maximal in its ambient space. Brouwer, Cohen and Neumaier, *Distance-Regular Graphs*, work with root lattices and do not use the notion.

Repair. The Lean definition and Definition 2.2 of the blueprint ask for maximality within $\mathrm{span}(\Phi)$. Lemma 2.3 is unchanged, since the sum of two roots lies in the span.

Status. Fixed. Blueprint: Definition 2.2, `def:2.2`, and Lemma 2.3, `lem:2.3` (`blueprint/sec2-lines.tex`); Theorem 3.6, `thm:3.6` (`blueprint/sec3-rootsystems.tex`). Lean: `LineSystem.IsMaximal` (`LineGraphs/LineSystem/Defs.lean`), with `LineSystem.IsMaximal.isStarClosed` for Lemma 2.3 and `LineSystem.classification_of_isMaximal` for Theorem 3.6.

Implicit hypothesis 8. Theorem 3.12: a part may be empty. *Where.* Theorem 3.12; p. 315.

Printed. The lines can be "partitioned into two subsets".

Missing. Either subset may be empty.

Evidence. With both subsets nonempty the theorem is false: a single line lies in a Root System but cannot be split into two nonempty parts, and two lines of A_2 at 60° must lie in the same part.

Repair. The partition is a colouring ϕ of S with two colours, not necessarily surjective, and the empty set of lines is allowed. The note `cgss/proofs/thm312/thm312.tex` proves the theorem in this form.

Status. Fixed. Blueprint: Theorem 3.12, `thm:3.12`, with its paragraph on the conventions (`blueprint/sec3-thm312.tex`), after `cgss/proofs/thm312/thm312.tex`. Lean: `LineSystem.subset_rootSet_iff_bipartition` (`LineGraphs/Theorem312/Main.lean`), whose colouring, in `LineSystem.HasBipartition`, takes values in `Bool`; its docstring says that the set and either colour fibre may be empty.